

Government College of Engineering Aurangabad, Chhatrapati Sambhajanagar

(An Autonomous Institute of Government of Maharashtra)

End Semester Examination April-May 2025

First Year MCA – NEP – Semester – II

MCPEC1002 Soft Computing

Time: 2.30 Hours

20 MAY 2025

Max. Marks: 60

“Verify the Course Code and check whether you have got the Correct Question Paper”

N.B.

1. Attempt all questions.

2. Figures to the right indicate full marks.

3. Assume suitable data if necessary and state it clearly.

4. Use of Non-Programmable Calculator and Data Sheet is allowed.

Q.1	Attempt any two	CO	B.T. Level	Marks
A	What is learning? State and explain different categories of learning.	1	K2	06
R ₂ B	Explain different NNs architectures with suitable figure. R ₂	1	K2	06
R ₂ C	State and explain few applications of ANNs which are used to solve some real life problems.	1	K2	06
Q.2	Attempt any two			
R ₃ A	State and explain perceptron learning algorithm.	2	K2	06
R ₂ B	What non-linearly separable problem? Solve XOR problem [Assume suitable sample data]	2	K2	06
C	Implement AND function using perceptron networks for binary inputs and targets.	2	K3	06
Q.3	Attempt any two			
A	State and explain the error back-propagation learning algorithm	3	K3	06
B	Use Adaline network to train AND-NOT function with bipolar inputs and targets and perform two epochs of training for the same.	3	K3	06
C	Find the weights required to perform the following classification using perceptron network. The vectors (1, 1, 1, 1) and (-1,1,-1,-1) are belonging to the class (so have target value 1), vectors (1, 1, 1, -1) and (1, -1, -1, 1) are not belonging to the class (so have target value -1). Assume learning rate as 1 and initial weights as 0.	3	K2	06

Q.4	Attempt any two	CO	B.T. Level	Marks
A	What is competition? Draw flowchart for Mexican hat network.	4	K3	06
B	Explain the algorithm for learning vector quantization with suitable data.	4	K3	06
C	Consider a Kohonen self-organizing net with two cluster units and five input units. The weight vectors for the cluster units are given by: $W_1 = [1.0 \ 0.9 \ 0.7 \ 0.5 \ 0.3]$ $w_2 = [0.3 \ 0.5 \ 0.7 \ 0.9 \ 1.0]$ $x = [0.0 \ 0.5 \ 1.0 \ 0.5 \ 0.0]$. Learning rate of 0.25. Use the Euclidean distance to find the winning cluster unit & new weights for the winning unit.	4	K3	06
Q.5	Attempt any two			
A	Explain fuzzy sets with the properties and operations.	5	K2	06
B	For a give fuzzy sets A and B [Assume suitable members] perform Union, Intersection, complement and Minus operations.	5	K3	06
C	Use your own intuition and definition of the universe of discourse plot a member fuzzy function for weight of the people.	5	K3	06